
MODEL SELECTION AND VALIDATION REPORT (remove_3_biomass model)

OBSERVATIONS:1719

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_3_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	6.42	0.32
remove_3_tric	0.27	0.03
remove_3_teve	-0.31	0.02
sbt_mean	0.21	0.09
sbt_sd	-0.04	0.04
sst_sd	0.03	0.04
log_chla_mean	0.02	0.03
depth	0.08	0.04
log_sum_fishing	0.09	0.03

Dispersion parameter: 0.81

Spatiotemporal AR1 correlation (rho): 0.94

Matérn range: 361.61

Spatiotemporal marginal AR1 SD: 0.94

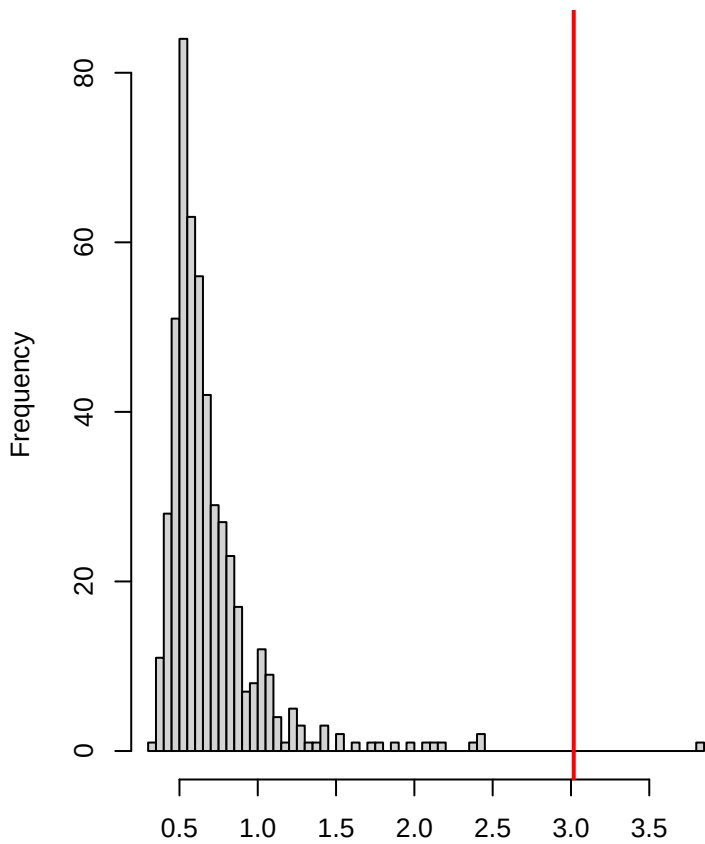
REML criterion at convergence: 12453.025

See ?tidy.sdmTMB to extract these values as a data frame.

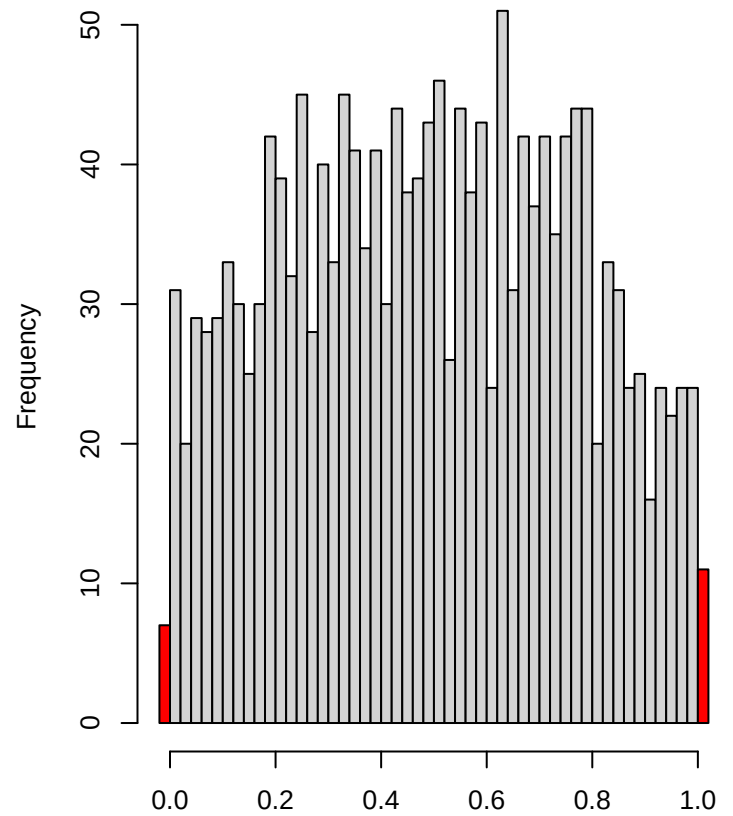
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	24932.05	2.964050
spatiotemporal_random_fields_ar1_with_anisotropy	15	24935.01	156.892887
spatial_random_fields	12	25091.91	2.141124
spatial_random_fields_with_anisotropy	14	25094.05	14.225348
spatiotemporal_random_fields_iid	12	25108.27	3.697342
spatiotemporal_random_fields_iid_with_anisotropy	14	25111.97	283.625053
error_only	10	25395.60	NA

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated

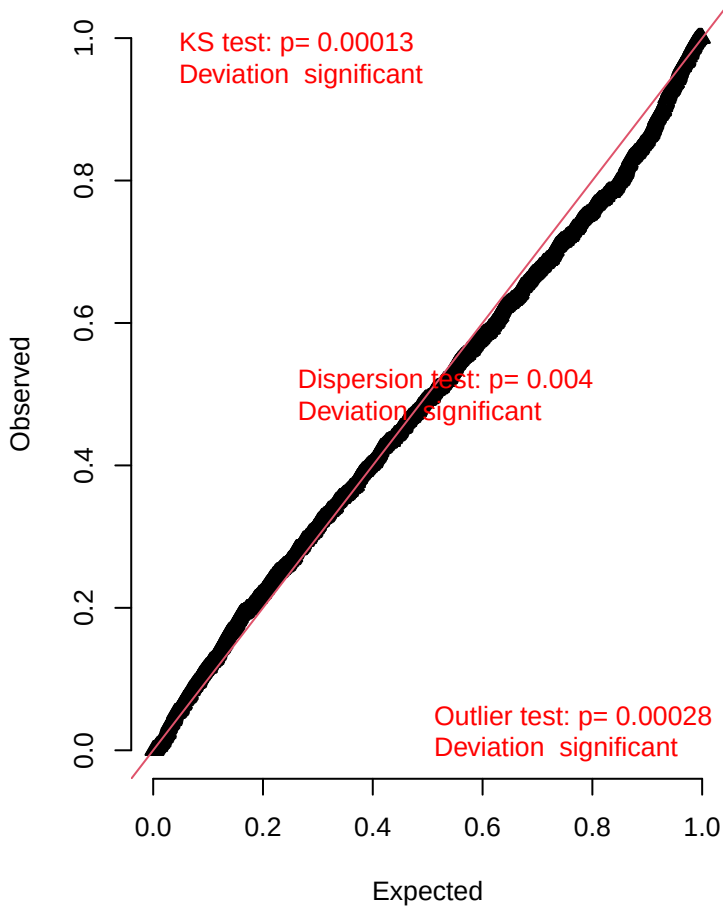


Outlier test significant



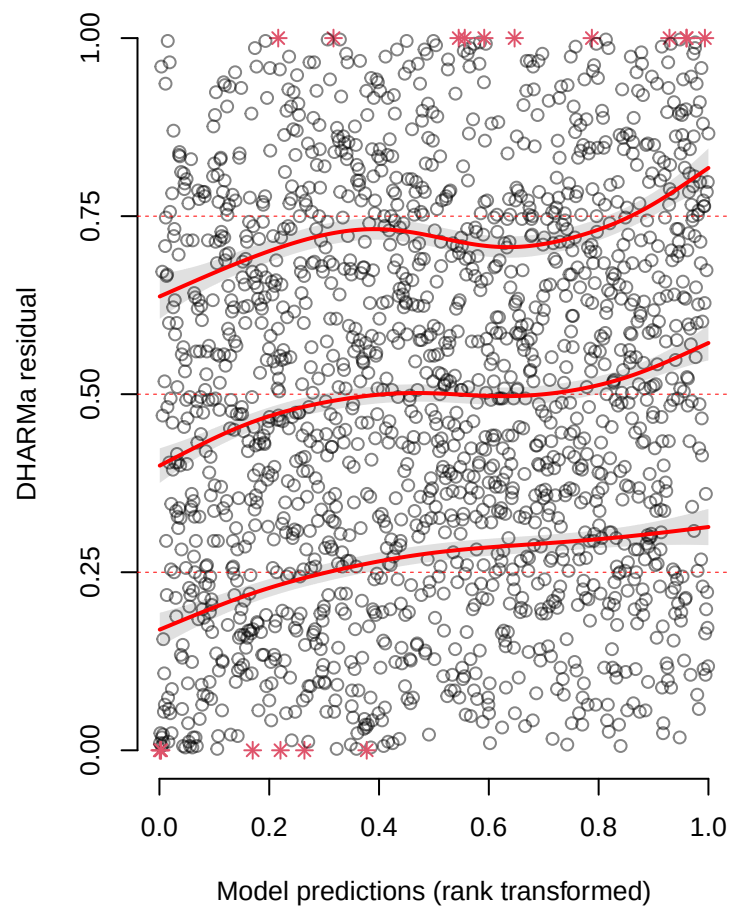
Simulated values, red line = fitted model. p-value (two.sided) =

QQ plot residuals

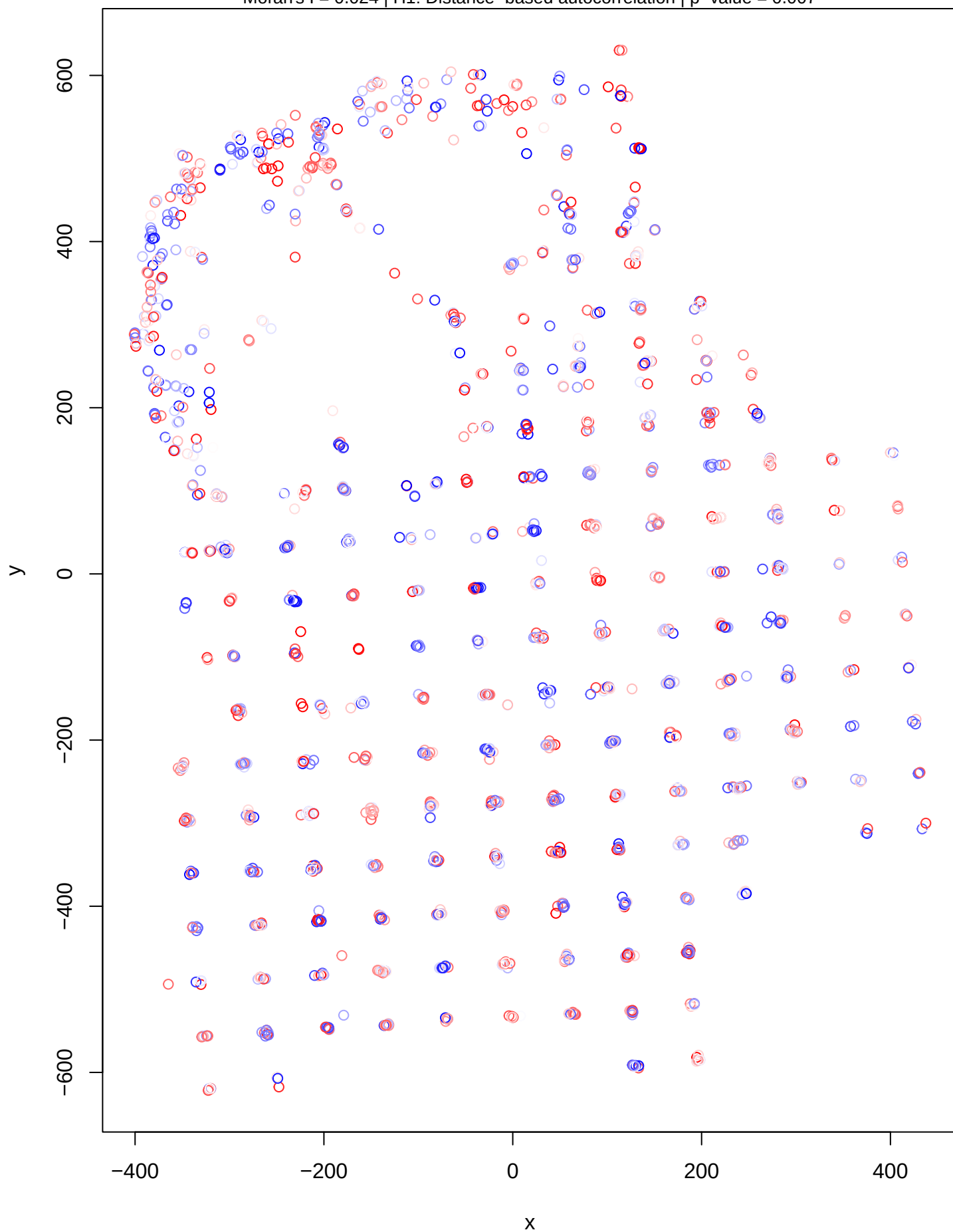


Residuals (outliers are marked red)

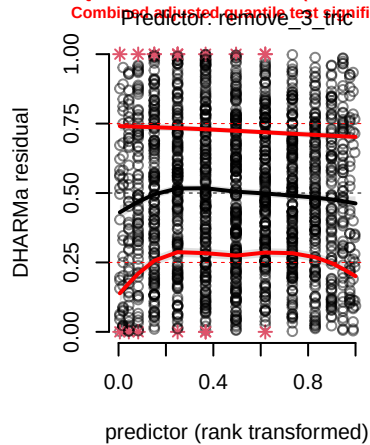
DHARMA residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



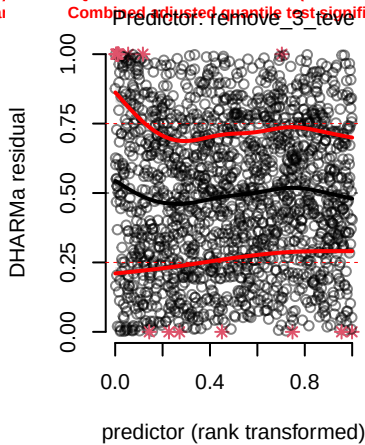
DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.024 | H1: Distance-based autocorrelation | p-value = 0.007



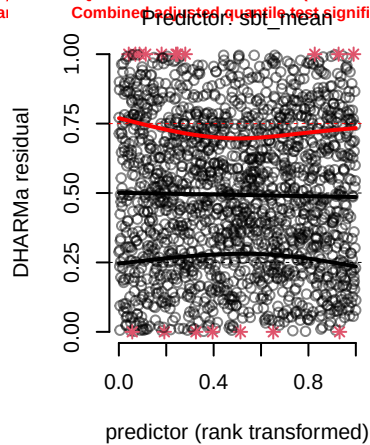
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



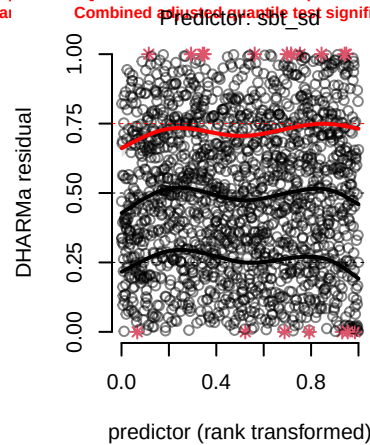
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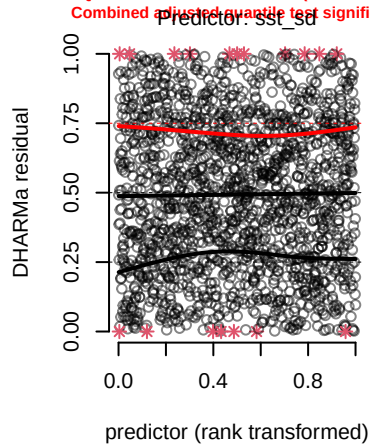
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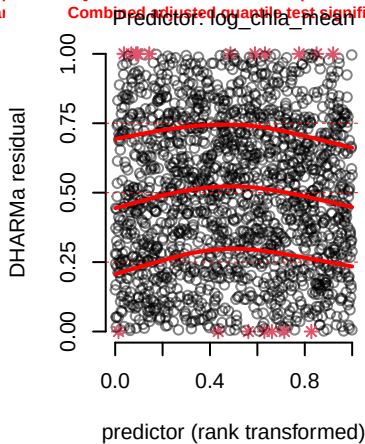
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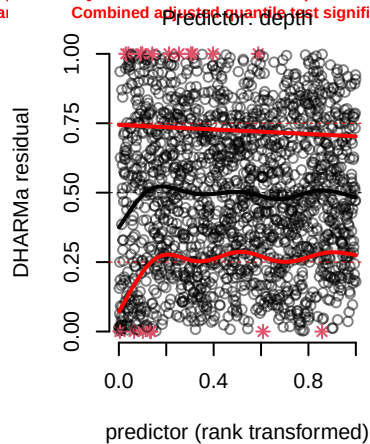
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